

We will cover only the following two topics in the Week 6 lecture:

- 1 Additivity of the double integral.
- 2 Double integrals in polar coordinates.

The Double Integral

Suppose that $D \subseteq \mathbb{R}^2$ is neither a Type I region nor a Type II region in $\mathbb{R}^2$.

Question: What methods do we have for computing a double integral over D ?

Answer: We can use the additivity of the double integral.

The Double Integral — Linearity

Theorem (Linearity of the Double Integral)

- Let D be a bounded region in $\mathbb{R}^2$ (not necessarily closed).
- Let f and g be real-valued functions defined on D .
- Let $c \in \mathbb{R}$.

If $\iint_D f(x, y) \, dA$ and $\iint_D g(x, y) \, dA$ exist, then the following statements hold:

- $\iint_D (f + g)(x, y) \, dA$ exists, and *sum*

$$T(u+v) = T(u) + T(v)$$

$$\iint_D (f + g)(x, y) \, dA = \iint_D f(x, y) \, dA + \iint_D g(x, y) \, dA.$$

- $\iint_D (c \cdot f)(x, y) \, dA$ exists, and

*similar to
linear transformation
in $\mathbb{R}^2$*

$$\iint_D (c \cdot f)(x, y) \, dA = c \iint_D f(x, y) \, dA.$$

The term “linearity” has the same meaning as it does in linear algebra.

The Double Integral — Integration over a Restricted Domain

Very often, when given a real-valued function f of two real variables whose domain is a bounded region of $\mathbb{R}^2$, we do not desire to calculate the double integral of f over all of $\text{Dom}(f)$ but rather a small portion of it.

How, then, shall we evaluate a double integral over a restricted domain?

Definition (Double Integrals over Restricted Domains)

- Let f be a real-valued function of two real variables.
- Suppose that $\text{Dom}(f)$ is a bounded region of $\mathbb{R}^2$.
- Let $D \subseteq \text{Dom}(f)$ — clearly, D is then a bounded region of $\mathbb{R}^2$ as well.

The **double integral of f over D** , denoted by $\iint_D f(x, y) \, dA$, is defined as follows:

$$\iint_D f(x, y) \, dA \stackrel{\text{df}}{=} \iint_D f|_D(x, y) \, dA.$$

Note: Here, “ $f|_D$ ” denotes the restriction of f to the possibly smaller domain D .

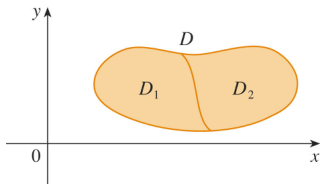
The Double Integral — Additivity

Theorem (Additivity of the Double Integral)

- Let $n \in \mathbb{N}$.
- Let $D_1, \dots, D_n$ be bounded regions in $\mathbb{R}^2$ (not necessarily closed).
- Let f be a real-valued function defined on the union $D \stackrel{\text{df}}{=} D_1 \cup \dots \cup D_n$.
- Suppose that the intersection of any two or more of $D_1, \dots, D_n$ has zero area.

If $\iint_{D_1} f(x, y) \, dA, \dots, \iint_{D_n} f(x, y) \, dA$ all exist, then $\iint_D f(x, y) \, dA$ exists, and

$$\iint_D f(x, y) \, dA = \iint_{D_1} f(x, y) \, dA + \dots + \iint_{D_n} f(x, y) \, dA.$$



The Double Integral — Integration over Non-Type-I/II Regions in $\mathbb{R}^2$

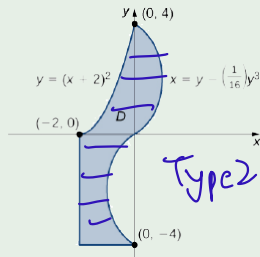
The theorem on the preceding slide yields an effective strategy for evaluating a double integral over a bounded region in $\mathbb{R}^2$ that is neither of Type I nor of Type II:

- Let D be a bounded region in $\mathbb{R}^2$ that is neither of Type I nor of Type II.
- However, suppose that D is the union of finitely many bounded regions in $\mathbb{R}^2$ that are either of Type I or of Type II.
- Then, provided that the intersection of any two or more of these regions has zero area, we can evaluate a double integral over D by summing a collection of double integrals over these regions, which we know how to evaluate.

The Double Integral — Example 1

Example

Let D be the bounded region of $\mathbb{R}^2$ depicted below:

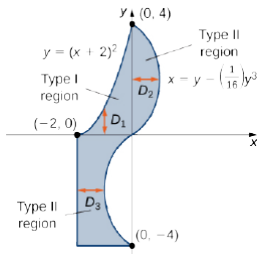


Problem: Evaluate $\iint_D (2x + 5y) \, dA$.

The Double Integral — Example 1 (cont'd)

Solution

- Let us first visualize D as the union of three bounded regions in $\mathbb{R}^2$ that are either of Type I or of Type II.



- Observe that D_1 is of Type I and that D_2 and D_3 are of Type II; specifically,

$$D_1 = \left\{ (x, y) \in \mathbb{R}^2 \mid -2 \leq x \leq 0 \text{ and } 0 \leq y \leq (x + 2)^2 \right\},$$

$$D_2 = \left\{ (x, y) \in \mathbb{R}^2 \mid 0 \leq y \leq 4 \text{ and } 0 \leq x \leq y - \frac{y^3}{16} \right\},$$

$$D_3 = \left\{ (x, y) \in \mathbb{R}^2 \mid -4 \leq y \leq 0 \text{ and } -2 \leq x \leq y - \frac{y^3}{16} \right\}.$$

The Double Integral — Example 1 (cont'd)

- Firstly, let us compute $\iint_{D_1} (2x + 5y) \, dA$:

$$\begin{aligned}\iint_{D_1} (2x + 5y) \, dA &= \int_{-2}^0 \left[\int_0^{(x+2)^2} (2x + 5y) \, dy \right] dx \\&= \int_{-2}^0 \left[2xy + \frac{5y^2}{2} \right] \Big|_{y=0}^{y=(x+2)^2} dx \\&= \int_{-2}^0 \left[2x(x+2)^2 + \frac{5(x+2)^4}{2} \right] dx \\&= \int_{-2}^0 \left(\frac{5x^4}{2} + 22x^3 + 68x^2 + 88x + 40 \right) dx \\&= \left[\frac{x^5}{2} + \frac{11x^4}{2} + \frac{68x^3}{3} + 44x^2 + 40x \right] \Big|_{-2}^0 \\&= \frac{40}{3}.\end{aligned}$$

The Double Integral — Example 1 (cont'd)

- Next, let us compute $\iint_{D_2} (2x + 5y) \, dA$:

$$\begin{aligned}\iint_{D_2} (2x + 5y) \, dA &= \int_0^4 \left[\int_0^{y - \frac{y^3}{16}} (2x + 5y) \, dx \right] dy \\&= \int_0^4 \left[x^2 + 5xy \right]_{x=0}^{x=y - \frac{y^3}{16}} dy \\&= \int_0^4 \left[\left(y - \frac{y^3}{16} \right)^2 + 5 \left(y - \frac{y^3}{16} \right) y \right] dy \\&= \int_0^4 \left(\frac{y^6}{256} - \frac{7y^4}{16} + 6y^2 \right) dy \\&= \left[\frac{y^7}{1792} - \frac{7y^5}{80} + 2y^3 \right]_0^4 \\&= \frac{1664}{35}.\end{aligned}$$

The Double Integral — Example 1 (cont'd)

- Finally, let us compute $\iint_{D_3} (2x + 5y) \, dA$:

$$\begin{aligned}\iint_{D_3} (2x + 5y) \, dA &= \int_{-4}^0 \left[\int_{-2}^{y - \frac{y^3}{16}} (2x + 5y) \, dx \right] dy \\&= \int_{-4}^0 \left[x^2 + 5xy \right] \Big|_{x=-2}^{x=y - \frac{y^3}{16}} dy \\&= \int_{-4}^0 \left[\left[\left(y - \frac{y^3}{16} \right)^2 + 5 \left(y - \frac{y^3}{16} \right) y \right] - (4 - 10y) \right] dy \\&= \int_{-4}^0 \left(\frac{y^6}{256} - \frac{7y^4}{16} + 6y^2 + 10y - 4 \right) dy \\&= \left[\frac{y^7}{1792} - \frac{7y^5}{80} + 2y^3 + 5y^2 - 4y \right] \Big|_{-4}^0 \\&= -\frac{1696}{35}.\end{aligned}$$

The Double Integral — Example 1 (cont'd)

- The intersection of any two or more of D_1 , D_2 , and D_3 has zero area:
 - $D_1 \cap D_2 = \{0\} \times [0, 4]$ is a line segment in $\mathbb{R}^2$, which has zero area.
 - $D_1 \cap D_3 = [-2, 0] \times \{0\}$ is a line segment in $\mathbb{R}^2$, which has zero area.
 - $D_2 \cap D_3 = \{(0, 0)\}$ is a point in $\mathbb{R}^2$, which has zero area.
 - $D_1 \cap D_2 \cap D_3 = \{(0, 0)\}$ is a point in $\mathbb{R}^2$, which has zero area.
- Therefore, we can apply the additivity of the double integral to conclude that

$$\begin{aligned} & \iint_D (2x + 5y) \, dA \\ &= \iint_{D_1} (2x + 5y) \, dA + \iint_{D_2} (2x + 5y) \, dA + \iint_{D_3} (2x + 5y) \, dA \\ &= \frac{40}{3} + \frac{1664}{35} - \frac{1696}{35} \\ &= \frac{1304}{105} \\ &= 12 \frac{44}{105}. \end{aligned}$$

Double Integrals in Polar Coordinates — Polar Rectangles

Definition (the Polar Plane)

The **polar plane** is just the set $\mathbb{R}_{\geq 0} \times \mathbb{R}$.

Given a point (r, θ) in the polar plane, we can define the point

$$(x, y) \stackrel{\text{df}}{=} (r \cos(\theta), r \sin(\theta))$$

in the Cartesian plane $\mathbb{R}^2$, called the “**Cartesian image**” of (r, θ) .

each point will always have more than one set of polar coordinates.

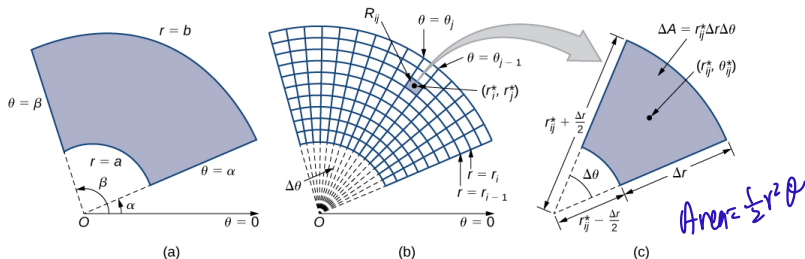
Conversely, given a point (x, y) in the Cartesian plane $\mathbb{R}^2$, we can find a point (r, θ) in the polar plane such that $(x, y) = (r \cos(\theta), r \sin(\theta))$, called a “**polar-coordinate pair**”.

Definition (Polar Rectangles)

A **polar rectangle** is a region D in $\mathbb{R}^2$ (necessarily bounded and closed) for which there exist

- $a, b \in \mathbb{R}_{\geq 0}$ with $a \leq b$, called the “**inner and outer radii**”, respectively, and
 - $\alpha, \beta \in \mathbb{R}$ with $0 \leq \beta - \alpha \leq 2\pi$, called the “**initial and final angles**”, respectively,
- such that D is the Cartesian image of the closed standard rectangle $[a, b] \times [\alpha, \beta]$ that is situated in the polar plane.

Double Integrals in Polar Coordinates — Integration over Polar Rectangles



Let $a, b \in \mathbb{R}_{\geq 0}$ and $\alpha, \beta \in \mathbb{R}$ satisfy $a \leq b$ and $0 \leq \beta - \alpha \leq 2\pi$.

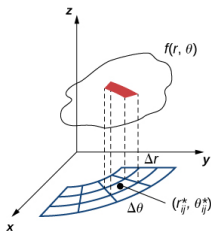
Figure (a) depicts the polar rectangle R specified by

$$\left\{ (r \cos(\theta), r \sin(\theta)) \in \mathbb{R}^2 \mid (r, \theta) \in [a, b] \times [\alpha, \beta] \right\}.$$

Figure (b) depicts a division of R into polar sub-rectangles; each polar sub-rectangle is the Cartesian image of a closed standard sub-rectangle of $[a, b] \times [\alpha, \beta]$.

Figure (c) says that if the center of the (i, j) -th polar sub-rectangle $R_{i,j}$ is the Cartesian image of $(r_{i,j}^*, \theta_{i,j}^*) \in [a, b] \times [\alpha, \beta]$, then the area $(\Delta A)_{i,j}$ of $R_{i,j}$ is $r_{i,j}^* \cdot (\Delta r)(\Delta \theta)$.

Double Integrals in Polar Coordinates — Integration over Polar Rectangles



Let f be a continuous, non-negative, real-valued function defined on R .

We can approximate the volume between $R \times \{0\}$ and the graph of f by the sum

$$\begin{aligned} & \sum_{i=1}^m \sum_{j=1}^n f(r_{i,j}^* \cos(\theta_{i,j}^*), r_{i,j}^* \sin(\theta_{i,j}^*)) \cdot (\Delta A)_{i,j} \\ &= \sum_{i=1}^m \sum_{j=1}^n f(r_{i,j}^* \cos(\theta_{i,j}^*), r_{i,j}^* \sin(\theta_{i,j}^*)) \cdot r_{i,j}^* \cdot (\Delta r)(\Delta \theta). \end{aligned}$$

$\Delta x \Delta y$
 $\Rightarrow dA(r, \theta)$

This is a Riemann sum, so if we let $m, n \rightarrow \infty$, we get the theorem on the next slide.

Theorem (Double Integrals over Polar Rectangles)

- Let $a, b \in \mathbb{R}_{\geq 0}$ and $\alpha, \beta \in \mathbb{R}$ satisfy $a \leq b$ and $0 \leq \beta - \alpha \leq 2\pi$.
- Let R denote the polar rectangle that is the Cartesian image of $[a, b] \times [\alpha, \beta]$.
- Let f be a continuous real-valued function defined on R .

Then $\iint_R f(x, y) \, dA(x, y)$ exists, and, by Fubini's Theorem, we have

$$\begin{aligned} \iint_R f(x, y) \, dA(x, y) &= \iint_{[a, b] \times [\alpha, \beta]} f(r \cos(\theta), r \sin(\theta)) \cdot r \, dA(r, \theta) \\ &= \int_a^b \left[\int_\alpha^\beta f(r \cos(\theta), r \sin(\theta)) \cdot r \, d\theta \right] dr \\ &= \int_\alpha^\beta \left[\int_a^b f(r \cos(\theta), r \sin(\theta)) \cdot r \, dr \right] d\theta. \end{aligned}$$

This theorem allows the evaluation of double integrals over bounded regions in $\mathbb{R}^2$ that may not be of Type I or Type II but are at least polar rectangles!

Double Integrals in Polar Coordinates — Example 1

Example

Let D denote the annular region in the Cartesian plane $\mathbb{R}^2$ given by

$$\{(x, y) \in \mathbb{R}^2 \mid x \leq 0 \text{ and } 1 \leq x^2 + y^2 \leq 4\}.$$

Problem: Evaluate $\iint_D (x + y) \, dA$.

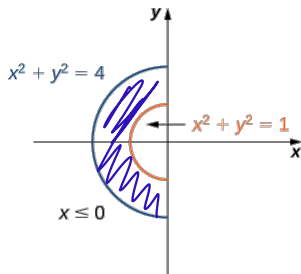
↓
left of y-axis

↓
hint circle!

usually have
to sketch!
got used to it

Solution

- Let us first visualize D with the aid of a figure.



inner radius = 1
outer radius = 2

Double Integrals in Polar Coordinates — Example 1 (cont'd)

- Hence, D is a polar rectangle that is the Cartesian image of $[1, 2] \times \left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$.
- Therefore,

$$\begin{aligned}\iint_D (x + y) \, dA &= \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \left[\int_1^2 [r \cos(\theta) + r \sin(\theta)] \cdot r \, dr \right] d\theta \\&= \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \left[\int_1^2 r^2 [\cos(\theta) + \sin(\theta)] \, dr \right] d\theta \\&= \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \left[[\cos(\theta) + \sin(\theta)] \cdot \left(\int_1^2 r^2 \, dr \right) \right] d\theta \\&= \left[\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} [\cos(\theta) + \sin(\theta)] \, d\theta \right] \cdot \left(\int_1^2 r^2 \, dr \right) \\&= [\sin(\theta) - \cos(\theta)] \Big|_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \cdot \left[\frac{r^3}{3} \right]_1^2 \\&= -2 \cdot \frac{7}{3} \\&= -\frac{14}{3}.\end{aligned}$$

$$\iint f(x)g(y) \, dA$$

$$\left(\int f(x) \, dx \right) \left(\int g(y) \, dy \right)$$

Double Integrals in Polar Coordinates — Polar Regions

Definition (Polar Regions)

A **polar region** is a region D in $\mathbb{R}^2$ (automatically bounded and closed) for which there exist

- $\alpha, \beta \in \mathbb{R}$ with $0 \leq \beta - \alpha \leq 2\pi$, and
- continuous functions $h_1, h_2 : [\alpha, \beta] \rightarrow \mathbb{R}_{\geq 0}$ satisfying:
 - for all $\theta \in [\alpha, \beta]$, we have $h_1(\theta) \leq h_2(\theta)$, and
 - if $\beta - \alpha = 2\pi$, then $h_1(\alpha) = h_1(\beta)$ and $h_2(\alpha) = h_2(\beta)$;

h_1 - inner

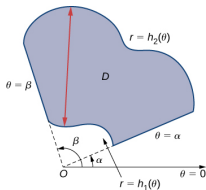
h_2 - outer

such that

$$D = \left\{ (r \cos(\theta), r \sin(\theta)) \in \mathbb{R}^2 \mid (r, \theta) \in [h_1(\theta), h_2(\theta)] \times [\alpha, \beta] \right\}.$$

depends on θ

Remark: Every polar rectangle is automatically a polar region.



Double Integrals in Polar Coordinates — Integration over Polar Regions

Theorem (Integration over Polar Regions)

- Let $\alpha, \beta \in \mathbb{R}$ satisfy $0 \leq \beta - \alpha \leq 2\pi$.
- Let h_1 and h_2 be continuous, non-negative, real-valued functions defined on $[\alpha, \beta]$ with the following two properties:
 - For all $\theta \in [\alpha, \beta]$, we have $h_1(\theta) \leq h_2(\theta)$.
 - If $\beta - \alpha = 2\pi$, then $h_1(\alpha) = h_1(\beta)$ and $h_2(\alpha) = h_2(\beta)$.
- Let D denote the polar region specified by

$$\left\{ (r \cos(\theta), r \sin(\theta)) \in \mathbb{R}^2 \mid (r, \theta) \in [h_1(\theta), h_2(\theta)] \times [\alpha, \beta] \right\}.$$

- Let f be a continuous real-valued function defined on D .

Then $\iint_D f(x, y) \, dA$ exists, and

$$\iint_D f(x, y) \, dA = \int_{\alpha}^{\beta} \left[\int_{h_1(\theta)}^{h_2(\theta)} f(r \cos(\theta), r \sin(\theta)) \cdot r \, dr \right] d\theta.$$

Don't forget to times r !
on the inside!
on the outside!

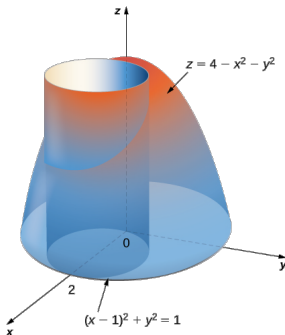
Example

Let D denote the closed disk in $\mathbb{R}^2$ given by the inequality $(x - 1)^2 + y^2 \leq 1$.

Problem: Find the volume V of the closed region in $\mathbb{R}^3$ under the paraboloid given by the equation $z = 4 - x^2 - y^2$ and above $D \times \{0\}$.

Solution

- Let us first visualize the closed region with the aid of a figure.



Double Integrals in Polar Coordinates — Example 2 (cont'd)

- Next, let us next specify the closed disk D in $\mathbb{R}^2$ as a polar region.
- Observe, for all $x, y \in \mathbb{R}$, that $(x - 1)^2 + y^2 \leq 1 \iff x^2 + y^2 \leq 2x$.
- A point in D is thus the Cartesian image of a point (r, θ) in the polar plane if and only if $r^2 \leq 2r \cos(\theta)$ if and only if $0 \leq r \leq 2 \cos(\theta)$.
- This allows us to specify D as a polar region in the following manner:

$$D = \left\{ (r \cos(\theta), r \sin(\theta)) \in \mathbb{R}^2 \mid (r, \theta) \in [0, 2 \cos(\theta)] \times \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \right\}.$$

Double Integrals in Polar Coordinates — Example 2 (cont'd)

- Therefore,

$$\begin{aligned} V &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left[\int_0^{2 \cos(\theta)} \left[4 - \overset{4 - x^2 - y^2}{[r \cos(\theta)]^2} - [r \sin(\theta)]^2 \right] \cdot \overset{\text{don't forget!}}{r} dr \right] d\theta \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left[\int_0^{2 \cos(\theta)} (4 - r^2) \cdot r dr \right] d\theta \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left[\int_0^{2 \cos(\theta)} (4r - r^3) dr \right] d\theta \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left[2r^2 - \frac{r^4}{4} \right] \Big|_{r=0}^{r=2 \cos(\theta)} d\theta \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} [8 \cos^2(\theta) - 4 \cos^4(\theta)] d\theta \\ &= \left[\frac{5\theta}{2} + \frac{5 \sin(\theta) \cos(\theta)}{2} - \sin(\theta) \cos^3(\theta) \right] \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \\ &= \frac{5\pi}{2}. \end{aligned}$$